# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

## CO2 AT1 — ANALYTICAL PROBLEM SOLVING

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## Question 1 — Subnetting 192.168.20.0/24

**Question:** Divide the block for Lab A (/26), Lab B (/27), and Lab C (/28). Find network address, valid host range, broadcast address, and usable hosts.

**Brief explanation:** A /26 has 64 addresses, /27 has 32, and /28 has 16. Usable hosts are found by subtracting the network and broadcast addresses (2). Allocate each subnet in order.

**Lab A (/26)**
Network: 192.168.20.0/26
Valid hosts: 192.168.20.1–192.168.20.62
Broadcast: 192.168.20.63
Usable hosts: 62

**Lab B (/27)**
Network: 192.168.20.64/27
Valid hosts: 192.168.20.65–192.168.20.94
Broadcast: 192.168.20.95
Usable hosts: 30

**Lab C (/28)**
Network: 192.168.20.96/28
Valid hosts: 192.168.20.97–192.168.20.110
Broadcast: 192.168.20.111
Usable hosts: 14

**Result:** 112 addresses are used in total (64 + 32 + 16). The next unused address is 192.168.20.112.

## Question 2 — Two systems with /26 mask

**Question:** System 1 is 192.168.1.130 and System 2 is 192.168.1.190. The subnet mask is 255.255.255.192.

**Brief explanation:** The block size is 256 − 192 = 64. Both addresses fall in the .128–.191 subnet.

**Answer:**
Network: 192.168.1.128/26
Valid hosts: 192.168.1.129–192.168.1.190
Broadcast: 192.168.1.191
Both systems are in the same subnet. Usable hosts: 62.

## Question 3 — Department subnet allocation

**Question:** Allocate subnets from 172.16.0.0/16 for HR (/25), Sales (/26), IT (/27), and Admin (/28).

**Brief explanation:** Assign the largest subnet first, then use the next correctly aligned network address for each department.

**HR (/25):** Network 172.16.0.0/25; hosts 172.16.0.1–172.16.0.126; broadcast 172.16.0.127; 126 usable hosts.

**Sales (/26):** Network 172.16.0.128/26; hosts 172.16.0.129–172.16.0.190; broadcast 172.16.0.191; 62 usable hosts.

**IT (/27):** Network 172.16.0.192/27; hosts 172.16.0.193–172.16.0.222; broadcast 172.16.0.223; 30 usable hosts.

**Admin (/28):** Network 172.16.0.224/28; hosts 172.16.0.225–172.16.0.238; broadcast 172.16.0.239; 14 usable hosts.

**Unused range:** Starts at 172.16.0.240 and continues through 172.16.255.255, excluding the allocated subnets.

## Question 4 — Subnetting 172.16.10.0/24 using /27

**Question:** Find the number of subnets and hosts per subnet; identify the subnet for 172.16.10.78 and check 172.16.10.95.

**Brief explanation:** Borrowed bits = 27 – 24 = 3, so there are $2^3 = 8$ subnets. Host bits = 32 – 27 = 5, so usable hosts are $2^{\blacksquare} - 2 = 30$. Block size = 256 – 224 = 32.

**Answer:**
172.16.10.78 belongs to subnet 172.16.10.64/27.
Network: 172.16.10.64
Valid hosts: 172.16.10.65–172.16.10.94
Broadcast: 172.16.10.95
172.16.10.95 is the broadcast address, so it cannot be assigned to a host.

## Question 5 — IPv6 address compression and prefix

**Question:** Given 2001:0db8:0000:0000:0000:ff00:0042:8329, compress it, expand it, and find the /64 prefix and possible host addresses.

**Brief explanation:** Remove leading zeros in each group and replace the longest run of zero groups with ::. A /64 prefix uses the first four groups for the network.

**Compressed address:** 2001:db8::ff00:42:8329

**Expanded address:** 2001:0db8:0000:0000:0000:ff00:0042:8329

**/64 network prefix:** 2001:db8::/64 (also written as 2001:db8:0:0::/64).

**Possible interface IDs:** The remaining 64 bits give $2^{\blacksquare\blacksquare}$ = 18,446,744,073,709,551,616 possible bit patterns.

**End of Assessment Tool 1**